

Curriculum Vitae of Yuxin Fan

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Department of Mathematics
The Chinese University of Hong Kong
Hong Kong SAR, China

ACADEMIC BACKGROUND

PhD Student in Mathematics since 08/2023
The Chinese University of Hong Kong
Advisor: Prof. Bangti Jin
Expected completion: June 2027

B.A. in Economics 09/2019–07/2023
School of Economics, Shanghai University of Finance and Economics, China
Graduated with **Highest Honors**, ranking 1st in the Honors Program (1/12) and 3rd overall among Economics majors (3/187).

RESEARCH INTERESTS

Inverse problems for partial differential equations, Bayesian nonparametric statistics, uncertainty quantification.

PUBLICATIONS / PREPRINTS

Y. Fan, J. Hong, and B. Jin.
Solving inverse acoustic obstacle scattering problem from phaseless far-field measurements using deep neural network surrogates.
Inverse Problems **41** (2025), no. 9, 095011. [Journal PDF](#)

Y. Fan and B. Jin.
On the contraction rate of the posterior distribution for nonlinear PDE parameter identification.
Submitted to *SIAM/ASA Journal on Uncertainty Quantification*. [arxiv:2601.17805](#)

ACADEMIC ACTIVITIES

Graduate Summer School in Pure Mathematics July–August 2022
Soochow University, Jiangsu, China
Successfully completed and passed graduate-level courses in *Algebra*, *Topology*, and *Real Analysis*, with final examinations.

Workshop on High Performance Computing July 2023
Tianyuan Mathematical Center in Central China, Wuhan, China
Topics included the *minimax rate in machine learning*.

TALKS

Contributed Talk, The 10th Young Scholar Symposium of the East Asia Section of IPIA April 2026

National Center for Theoretical Sciences, Taipei, Taiwan.

Contributed Talk, 15th National Conference on Inverse Problems, Imaging and Their Applications May 2026

Wenzhou-Kean University, Wenzhou, China.

TEACHING EXPERIENCE

Teaching Assistant

The Chinese University of Hong Kong (for Mathematics majors)

Undergraduate (Year 1): *University Mathematics*.

Undergraduate (Year 3): *Numerical Methods for Differential Equations*.

Master's level: *Econometric Principles and Data Analysis*.

Shanghai University of Finance and Economics

Undergraduate (Year 3): *Scientific Computations* (for Data Science majors).

Undergraduate (Year 3): *Corporate Finance* (for Finance majors).

SELECTED AWARDS

National Scholarship, Ministry of Education of China 2020–2021, 2021–2022

Awarded to top undergraduate students nationwide based on academic excellence.

First Prize, Chinese Mathematics Competitions for College Students (Shanghai Division) 2022, 2023

Provincial-level first prize in a nationwide mathematics competition.

Outstanding Graduate of Shanghai Municipality 2023

Awarded by the Shanghai Municipal Government to outstanding graduating students for academic excellence and overall achievement.